Zoaib Sihorwala

704-805-0669 | zoaibsihorwala@gmail.com | linkedin.com/in/zoaibsihorwala | github.com/zoaibs | U.S. Citizen

Education

Georgia Institute of Technology Expected May 2028
B.S./M.S. in Computer Science | *GPA: 4.0/4.0* *Atlanta, GA*
Coursework: *Artificial Intelligence, Machine Learning, Algorithms, Knowledge Based AI, Probability and Statistics*

Experience

Amazon Web Services May 2026 – August 2026
Software Development Engineer Intern *New York City, NY*

- **Automated the compilation pipeline** for the AWS CloudFormation resource specification document, transforming a **5-hour** manual verification process into a **30 second review** - a **99.84%** time reduction.
- **Engineered a serverless orchestration pipeline** using **Python, AWS Step Functions, and Lambda** to fetch, parse, and aggregate **1000s** of weekly schema updates across **11000+ AWS resources** into S3 buckets.
- **Architected an AI-powered schema parser** leveraging **Amazon Bedrock** to turn manual reviews of **8,000+ lines of nested JSON** into reviewing an intelligent, **one-page** semantic summary.
- **Proactively mitigated critical production risks**, enabling on-call engineers to identify and intercept **155 breaking schema changes** that would have resulted in high-severity **Sev-2 incidents**.

Intuit Credit Karma May 2025 – August 2025
Data Science Intern *Charlotte, NC*

- **Built and processed a 22GB dataset** of user features, logins, notification history, and time-domain data using **BigQuery, SQL** and 6 DAGs in **Apache Airflow.**
- Used **Scikit-learn** to train a random forest regression model on the dataset to **generate 210000+ synthetic counterfactual labels**, estimating user logins had no notification been sent.
- Designed, built, and trained a **Tensorflow-based causal inference model** on the synthetic data and implemented an importance sampling based inference algorithm for accurate predictions.
- **Improved attribution rate by 52%** by scoring model on notification-attributed logins and identifying that less than half were validly attributed, potentially **saving costs for over 15 million notifications** daily.

Michigan State University June 2024 – August 2024
AI Research Intern *Lansing, MI*

- **Created a 40GB dataset of over 7500 high-resolution satellite images** along with meteorological and air quality data using Python-based geospatial analysis software and 3 weather APIs.
- Led the development of an ensemble computer vision model with Pytorch using transformers, convolutional neural networks, and several image processing techniques to predict Air Quality Index in polluted areas.
- Trained model using parallel processing on MSU's **HPCC supercomputer and eight GPU servers.**
- **Achieved testing accuracy of 94%**, superior to existing state-of-the-art computer vision models.

Projects

Dynamic KV Cache Scheduler | *PyTorch, OpenCV, Vision Transformers* May 2026 – Present

- Developed a **hybrid dynamic KV quantization framework** for transformer-based video models that adaptively adjusts cache bit precision based on scene complexity and motion intensity.
- Engineered an OpenCV-powered "information score" pipeline to drive real time precision scheduling, reducing **KV cache memory footprint by 32%** while maintaining within 1.2% of full-precision accuracy.
- Integrated rotation-based outlier mitigation and sink-token preservation techniques to stabilize low bit inference, achieving a **14% throughput improvement** and **18% VRAM reduction** compared to uniform baselines.

MemoirAI | *Python, LangChain, OpenAI CLIP, VectorDB* August 2024 – December 2024

- Engineered a query optimization agent using **LangChain and GPT-4o**, which leveraged NLP to convert user inputs into vector search queries, achieving an average **processing time of 0.6 seconds.**
- Implemented an image analysis agent using the **OpenAI CLIP model** to automate the conversion of **1500+** images into vector embeddings, building the foundational semantic index stored in a Pinecone vector database.

Technical Skills

Languages: Java, Python, SQL, C, C++, C#, HTML, CSS, Javascript
Tools: PyTorch, TensorFlow, scikit-learn, BigQuery, Git (Version Control), Pandas, Github Copilot, Apache Airflow